

choptuik_ac_bc · theorems12 · github.com/wild8highlander/choptuik_ac_bc

The Second-Flows Theorem: the limit-cycle variables carry no genuine linear dynamics

char degree 22 = $9\lambda^4 \cdot Q18$ · 40-digit verdicts · 0 genuine at baseline, 1 on the brick

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Status:	PROVED EXACTLY (char factorization) + 40-digit filter

Abstract

This supplementary monograph records the second-flows theorem: the limit-cycle variables of the corrected tower carry no genuine linear dynamics. The exact characteristic polynomial of the second-flow quadratic eigenvalue problem has degree 22 and factors with the head $9\lambda^4 \cdot Q18(\lambda)$; the multiplicity of $\lambda = 0$ is exactly four, the exact rank of the constraint matrix C is 3, and the 40-digit verification (Newton + mpmath) confirms every QEP root: at the baseline the only borderline spectral point is $\lambda = 0$ (kernel dimension 1), all other modes are phantoms ($n_{\text{genuine}} = 0$).

On the one-brick point exactly one genuine mode appears — the clock root itself ($\tau^* = 7.1079202069\dots$); the second-flow sector inherits the clock and adds nothing. The V1 kernel at $\lambda = 0$ is one-dimensional at both points (rel σ_{min} at zero $\approx 10^{-18}$ – 10^{-19}). Tables of the exact factorization and the 40-digit verdicts, corollaries, honest caveats and the reproduction protocol are included.

Keywords: second flows, limit-cycle variables, QEP, char degree 22, 40-digit verification, phantoms, SymPy, machine theorem.

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1. Introduction

This supplementary monograph records the second-flows theorem — the machine result of campaign v13 (machines `second_flows_exact.py` and `second_flows_verify.py`, verdicts `second_flows_exact.json` and `second_flows_verified.json`, LEDGER §24). The second flows were introduced as limit-cycle variables — coordinates adapted to the would-be oscillatory carrier — and the question was whether they carry genuine linear dynamics that the amplitude sector missed.

The answer is negative and machine-rigorous. The exact characteristic polynomial of the second-flow QEP factors with the head $9\lambda^4$, the fourfold zero is exact, and the 40-digit Newton/mpmath filter classifies every root: nothing genuine at the baseline, the clock root alone on the brick. The theorem completes the linear annihilation chain: the weight rule (No. 4), the dd-closure (No. 5) and now the second flows agree that no linear mechanism produces the spinor structure — it must be finite-amplitude.

2. Notation and Setting

The second-flow variables (W2, R3, P4 as limit-cycle coordinates) linearize into a quadratic eigenvalue problem $\text{QEP}(\lambda)$ of dimension 22; the pencil C couples the second flows to the tower amplitudes. The working precision of the verification layer is 40 digits (mpmath); the Newton polishing refines each floating-point root and re-evaluates the relative singular minimum $\sigma_{\min}/\|M\|$ at the polished point.

The classification thresholds: $\text{rel_after_mp} \leq 10^{-9}$ with a clean σ gap — GENUINE or BORDERLINE (at the machine zero); $\text{rel_after_mp} \geq 10^{-5}$ — PHANTOM. The two convention points of the programme (baseline $\kappa = 2$, one-brick $\kappa = 2 - \pi^2/98$) are both processed.

3. Statement of the Theorem

Theorem 15 (the linear verdict on the second flows). For the corrected tower:

- (i) the exact characteristic polynomial of the second-flow QEP has degree 22 and factors with the head $9\lambda^4 \cdot Q_{18}(\lambda)$; the multiplicity of $\lambda = 0$ is exactly 4 (`multiplicity_lambda0 = 4`), the crosscheck at $\lambda = 1/2$ holds (`crosscheck_lam_half = true`), and the exact rank of C is 3 (`rank_C_exact = 3`);
- (ii) at the baseline the 40-digit verification classifies all roots: one BORDERLINE point at $\lambda = 0$ (`rel_after_mp = 4.4 \times 10^{-21}`, `$\sigma_{\min_{\text{abs}}} = 3.94 \times 10^{-18}$`) and every other mode a PHANTOM (`rel_after_mp  $\approx 10^{-9}$` with `$\sigma_{\min}/\sigma_{\max} \approx 10^{-6}$`); `n_genuine = 0`;
- (iii) the V1 kernel of the second-flow block at $\lambda = 0$ is one-dimensional at both points (`rel  $\sigma_{\min}$  at 0:  $4.42 \times 10^{-18}$  baseline,  $2.50 \times 10^{-19}$  brick; kernel_dim_approx = 1);`
- (iv) on the one-brick point exactly one genuine mode appears: the clock root $\tau^* = 7.1079202069226305031$ — the second-flow sector inherits the brick clock and adds no new linear scales (`n_genuine = 1`).

4. Sketch of the Machine Proof

The exact layer builds the QEP symbolically over $\mathbb{Q}(\sqrt{3})$, computes the characteristic polynomial, verifies the factor head and the $\lambda = 0$ multiplicity by exact differentiation, and checks the rank of C by exact elimination. The verification layer takes every floating-point root of the QEP, polishes it by Newton in 40-digit arithmetic, and evaluates the relative singular minimum at the polished point; the classification thresholds then sort the roots. The two layers agree on every root — the exact factorization predicts which roots are structural zeros, and the filter confirms it numerically.

Table 15.1. Layers of the two verdicts

Layer	Statement	Status
exact char	degree 22; head $9\lambda^4 \cdot Q_{18}$; mult(0) = 4; rank $C = 3$	exact ($\mathbb{Q}(\sqrt{3})$)
40d baseline	1 borderline at 0; all others phantoms; <code>n_genuine = 0</code>	mpmath 40 digits
40d brick	1 genuine = the clock root $7.1079202069\dots$; <code>n_genuine = 1</code>	mpmath 40 digits

Layer	Statement	Status
V1 kernel	$\dim \approx 1$ at $\lambda = 0$, both points (σ at $0 \sim 10^{-18}$ – 10^{-19})	numerical

5. Numerical Results

Table 15.2. The 40-digit verdicts at the baseline (representative roots)

λ (polished)	rel $\sigma_{\min}$ (40d)	Verdict
$3.96 \times 10^{-22} + 0i$	4.40×10^{-21}	BORDERLINE ($\lambda = 0$)
$-2.12 \times 10^{-5} + 0.00187i$	1.40×10^{-9}	PHANTOM
$0.00193 + 1.34 \times 10^{-5}i$	1.51×10^{-9}	PHANTOM
$-7.58 - 4.31i$	4.46×10^{-4}	PHANTOM
$-6.98 \pm 0.43i$	7.79×10^{-5}	PHANTOM

The one-brick table carries the single genuine point: $\lambda(T0h)$ with $T0h = 2.6660683\dots$, $\tau^* = 7.1079202069226305031$ — the same number the clock-pair theorem (No. 1) and the ladder (No. 2) deliver. The second-flow sector is therefore clock-transparent: it reproduces the clock and contributes no linear structure of its own.

6. Corollaries

Corollary 15.1 (the limit-cycle reading is empty at the linear level). The variables designed to carry an oscillatory carrier carry none: their spectrum is the fourfold zero plus phantoms. Any limit-cycle physics must be finite-amplitude — consistent with the flat-line theorem (No. 6) and the exit rates of the figure cycle (No. 14).

Corollary 15.2 (the clock propagates through all linear sectors). The brick clock appears as the only genuine second-flow mode: the clock is the invariant of the whole linear programme, now verified through a fourth independent sector (after the pair, the ladder, and the holonomy family).

Corollary 15.3 (the phantom filter is reusable). The exact-plus-40-digit protocol — predict the structural zeros exactly, then filter — is the template that theorems No. 4 and No. 13 reuse; every future linear sector should be closed by the same two-layer procedure.

7. Honest Caveats

The classification thresholds (10^{-9} genuine, 10^{-5} phantom) are filter conventions; the borderline label at $\lambda = 0$ records the structural fourfold zero rather than a marginal mode. The verification covers the two convention points; the brick-ladder theorem (No. 2) supplies the interpolation argument. The QEP is built for the truncated tower in the T1c source form; a tower enhancement changes the degree and requires a rebuild. Finally, the second flows are one linear sector among the tower's sectors; the theorem does not claim exhaustiveness over sectors not yet built.

8. Reproducibility

Table 15.3. Reproduction protocol for Theorem 15

Step	Command	Result
1	<code>python3 second_flows_exact.py</code>	<code>results/second_flows_exact.json</code> (char + 40d verdicts)
2	<code>python3</code> <code>second_flows_verify.py</code>	<code>results/</code> <code>second_flows_verified.json</code> (Newton + mpmath)
3	<code>python3 -m pytest tests/ -q</code>	58 passed (second-flow tests included)

Both machines run in seconds to minutes (deterministic, offline). The data correspond to commit 5b256cd; autonomous copies live in `einstein_direct/open9/machines/`.

References

- [1] Repository `wild8highlander/choptuik_ac_bc`: `einstein_direct/second_flows_exact.py`; `einstein_direct/results/second_flows_exact.json` (campaign v13, LEDGER §24).
- [2] `Ibid.`: `einstein_direct/second_flows_verify.py`; `einstein_direct/results/second_flows_verified.json` (40-digit verification).
- [3] Monographs No. 1, No. 2, No. 4, No. 5 of the theorems12 cycle.